

Hexagonal Constructor – Documentation

v1.1

1. Overview

Hexagonal Constructor is a procedural hex-grid generator for Unity featuring:

- Randomized generation (BranchWalk, ClusterGrowth)
 - Shape generation (Disk, Ring, Spiral)
 - Custom editor with preview and debugging tools
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2. Installation

1. Import the package through Package Manager.
 2. Add component **HexRoomGenerator.cs** to any GameObject.
 3. Assign a hex prefab in **Tile Visuals**.
 4. Adjust hex orientation in **Tile Geometry**.
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3. Inspector Summary

3. 1. Grid Settings

3. 1. 1. *Tile Visuals*

- **Hex Prefab** – prefab with SpriteRenderer
- **Hex Scale** – visual scale

3. 1. 2. *Tile Geometry*

- **Hex Orientation** – FlatTop / PointyTop
- **Hex Radius** – hex tile radius

3. 2. Generation Settings

- **Mode** – Randomized / Shapes
- **Start Axial** – starting hex tile coordinates (Q, R)

3. 2. 1. *Random Generation*

- **Random Algorithm** – BranchWalk / ClusterGrowth
- **Room Count** – tile count
- **Seed options** – deterministic generation

3. 2. 2. *Shape Generation*

- **Disk / Ring** – radius
 - **Spiral** – length, growth, start direction
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4. Editor Tools

4. 1. Editor Preview

- **Enable Preview** – toggles preview
- **Preview Color** – gizmo hexes color
- **Preview Hex Scale** – gizmo hex scale
- **Rebuild/Clear Preview** – generates gizmo hexes

4. 2. Generation Debug

- **Debug Mode** – toggles generator debug
 - **Step Delay** – time between step-by-step hex generation
 - **Rebuild/Clear Generation** – generates tiles in Scene View
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5. Notes

- Sprite rotation is manual for matching orientation
 - Grid logic is fully axial
 - Tested in Unity 2021.3+
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6. Troubleshooting

Tiles overlap → adjust radius / scale

Orientation looks wrong → rotate prefab (90°)